

ECSE 4660/6660: Internetworking of Things

Lecture 8: PHY Layer Fundamentals – Part VI

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ECSE, RPI

Spring 2026

Administrative

- Use Piazza for any questions
 - https://piazza.com/rpi/spring2026/ecs_e46606660
- Today's Quiz is available on **LMS**
 - Multiple choice questions
 - Due tonight 11:59 PM
- Project group formation
 - [IoT_SP26_project.xlsx](#)
- Project Proposal submission **due 2/9**
 - Template Provided
 - Rubrics Provided
 - Request extra office hour or visit my office JEC 6018 or post on Piazza
- HW1 on **LMS** **due 2/6 done**
 - No Late submission allowed
 - HW2 will be posted **due 2/20**
- Lab is located in JEC 5312
 - Access is provided to all registered students
 - You can visit lab anytime Mon-Fri 7AM-10PM
 - Labs are done **individually**- Plagiarism penalized heavily (See Syllabus pdf)
 - Late-submission Policy
 - **TA hours 4-5 PM** daily in 5312 lab
 - Lab 2 due **Fri, 2/13**

Why Measure and Model Networks

Ellen Zegura

Regents and Fleming Chair Professor, School of
Computer Science
Georgia Tech

Mercer Distinguished Lecture Series

LOW 3051

Wed, February 11, 2026 at 4:00 PM

In this talk, I will highlight three examples of network measurement and modeling projects that span my 30-year career. The first project advanced Internet topology modeling from simple random-graph based structures to incorporate hierarchy and locality, and spurred measurement efforts to better understand Internet structure. The second examined performance of the Domain Name System (DNS), a key service in the operation of the Internet since the mid-1980s. The third represents a five-year effort from research prototype to FCC-approved cellular measurement application and infrastructure, with the potential to impact mobile broadband investments. Throughout I will share lessons learned about research, development, funding, and working with students.

Dr. Ellen Zegura is Regents and Fleming Chair Professor in the School of Computer Science at Georgia Tech. In computer networking, she is known for her work on the GT-ITM suite of Internet topology tools, which were used for 20 years after initial release. In mobile wireless networking, she and colleagues invented the concept of message ferries to facilitate communications in environments where network connectivity is unreliable and/or sparse. Her work in computing and social good includes projects in Liberia with the Carter Center, with Native Americans in Southern California, and with residents of the Westside communities of Atlanta. She is currently on loan from Georgia Tech to the National Science Foundation where she serves as Acting Assistant Director for Computer and Information Science and Engineering (CISE). She served on the board of the Computing Research Association, including as Board Chair from 2019-2021. She is a Fellow of the IEEE and a Fellow of the ACM.

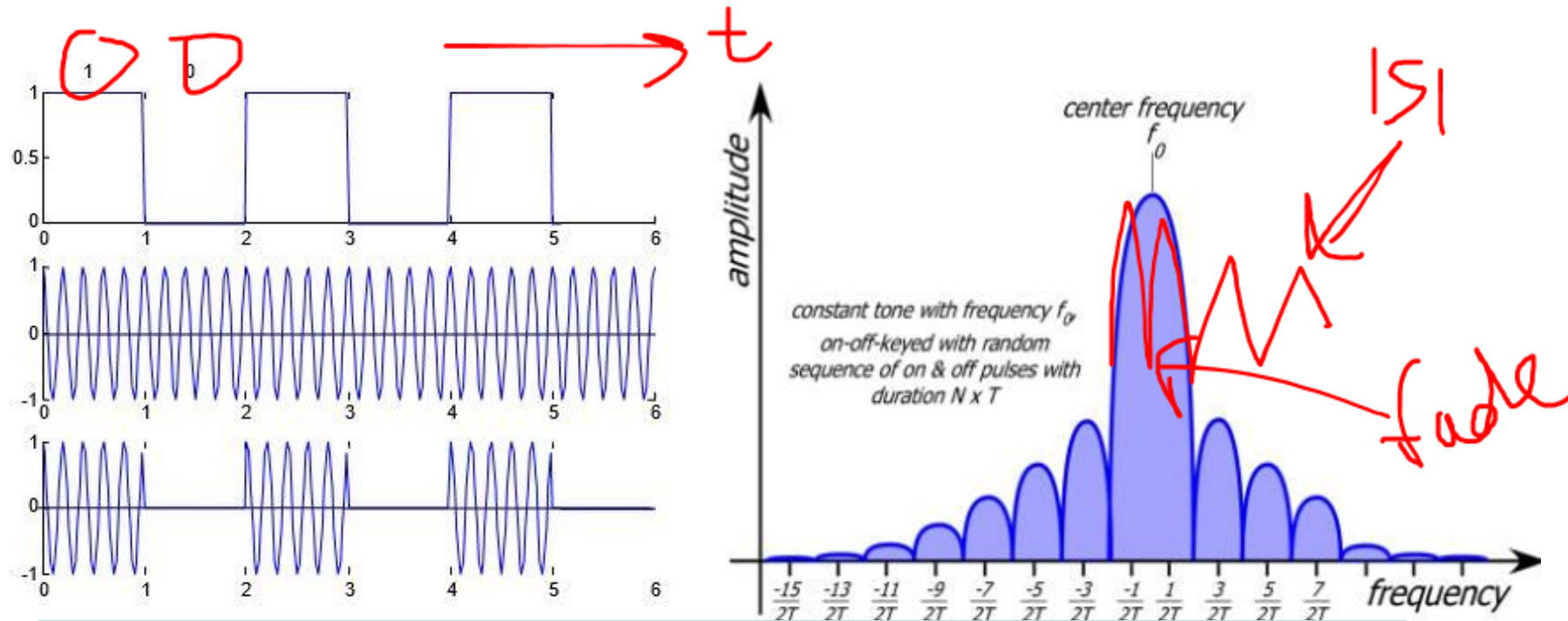


PHY Layer Techniques: Outline

1. OFDM (WiFi and Cellular)
 - High power, high rates, long range
2. Backscatter Communication and RFID
 - Extremely low-power, short range
3. Long Range (LoRa) Communication
 - Both low-power and long-range

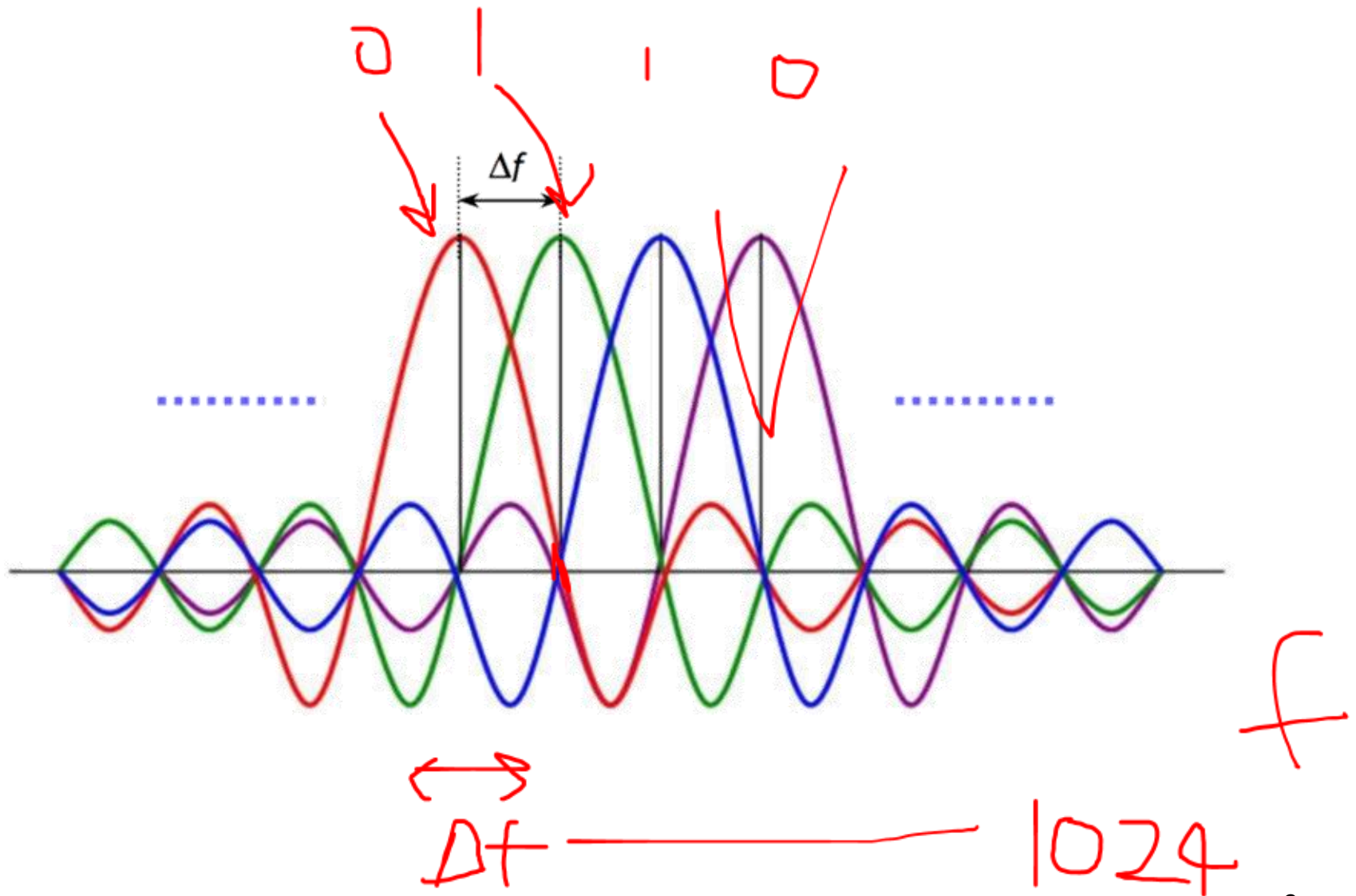
Digital Modulation

- Map Bits to Signal Values
- On-Off Keying (ASK):

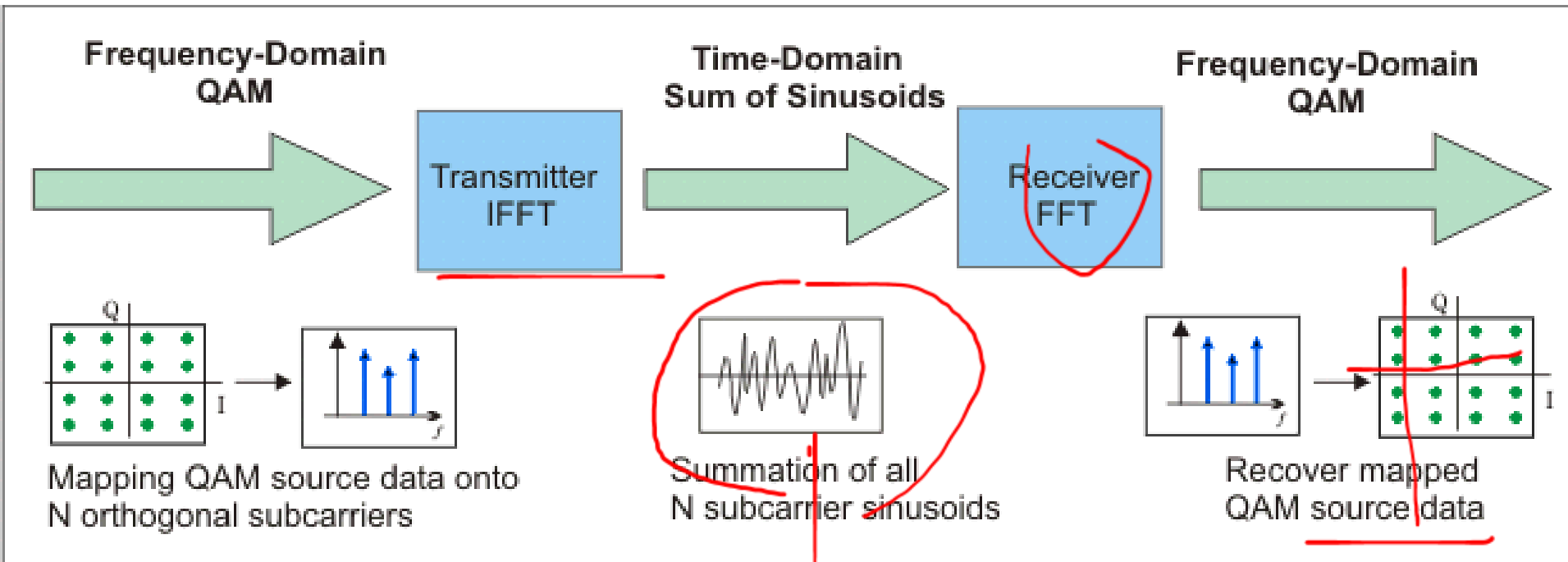


Suffers from Frequency-selective channel $\rightarrow$ ISI

OFDM Orthogonality



OFDM System block



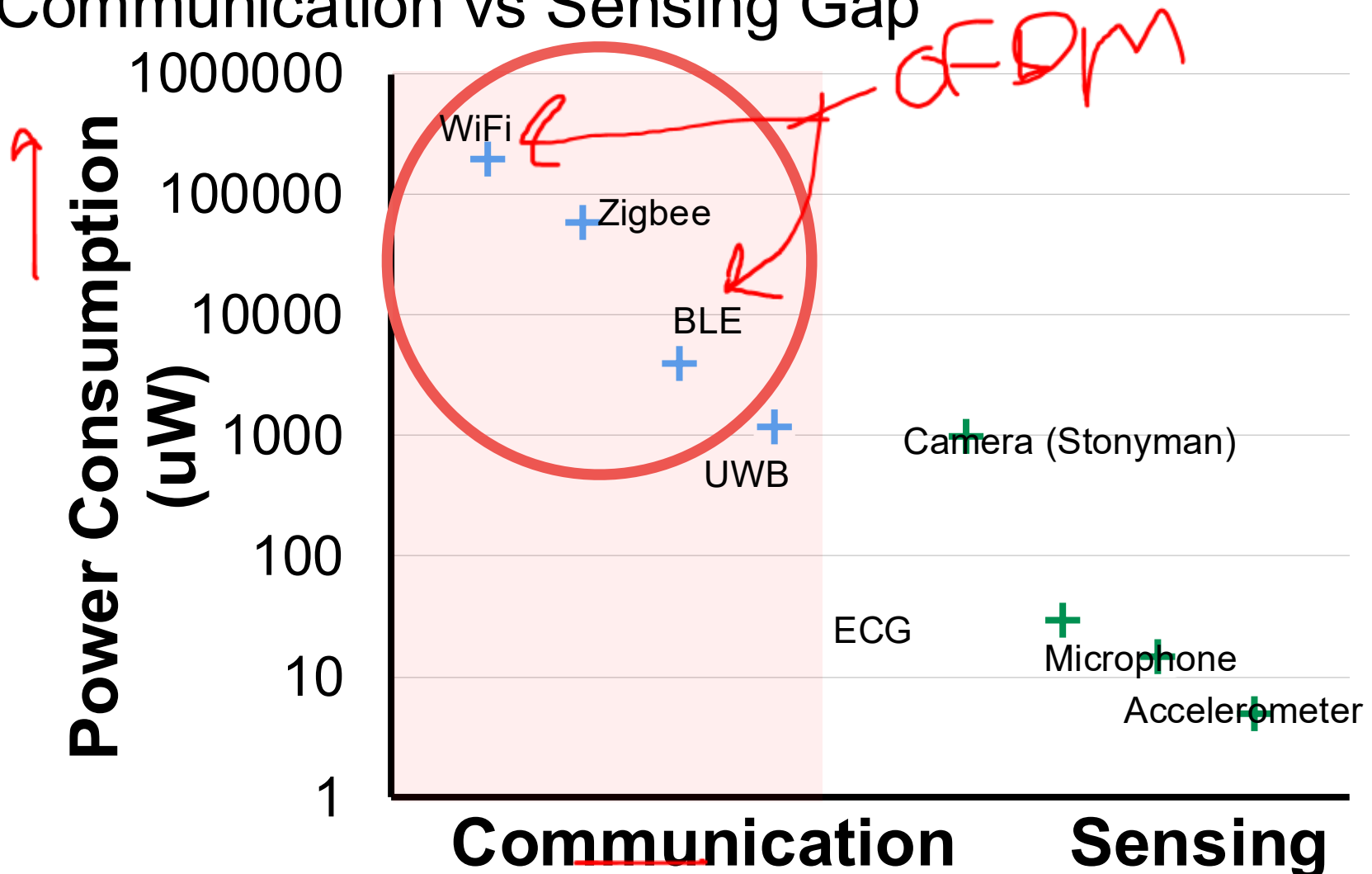
Simplified OFDM System Block Diagram

PAPR ↑

PHY Layer Techniques: Outline

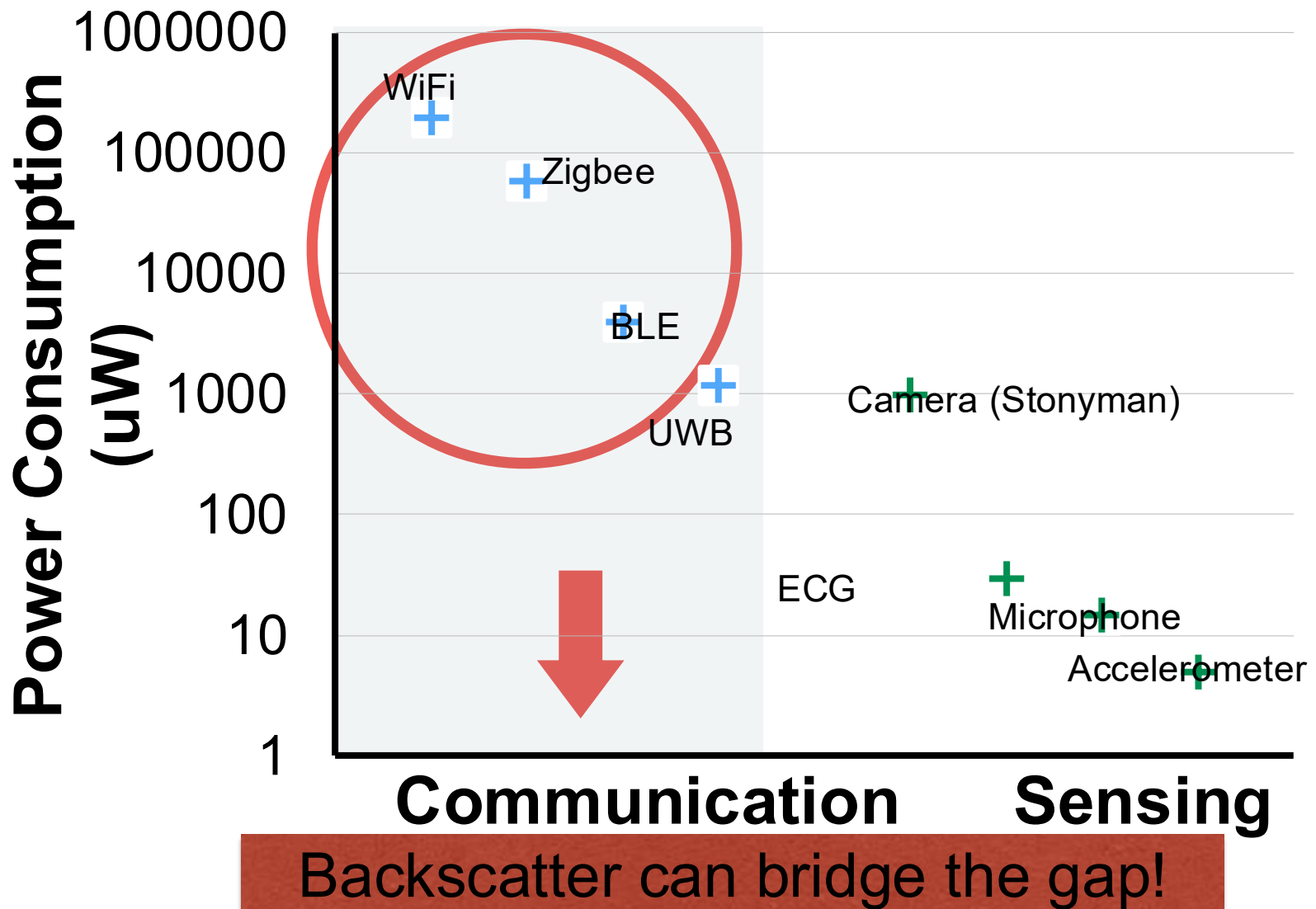
1. OFDM (WiFi and Cellular)
 - High power, high rates, long range
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 - **Extremely low-power, short range**
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Communication vs Sensing Gap

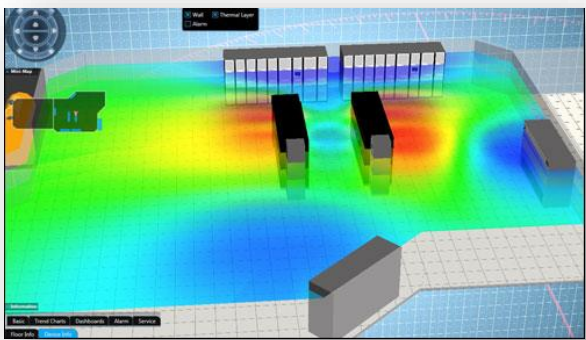
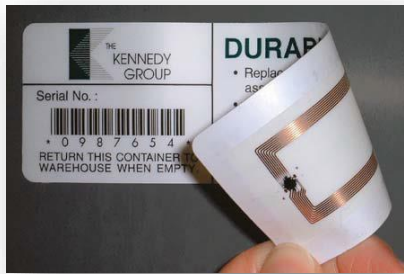


Radio: major bottleneck for low-power operation

A BIG Gap Between Radios and Sensors



RFIDs



Machine-Generated Data

RFID will be a major source of such traffic

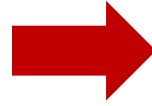
- In Oil & Gas – about 30% annual growth rate
- In Healthcare – \$1.3B revenue annually
- “number of RFID tags sold globally is projected to rise from 12 million in 2011 to **209 billion** in 2021.”
– *McKinsey Big Data Report 2011*

Can we use current wireless protocols for these low power networks?

NO

RFID Requirements

- Small form factor
- Massive scale
- Lifetime



RFID Constraints

- No battery
- Ultra-low cost
- Simple circuitry

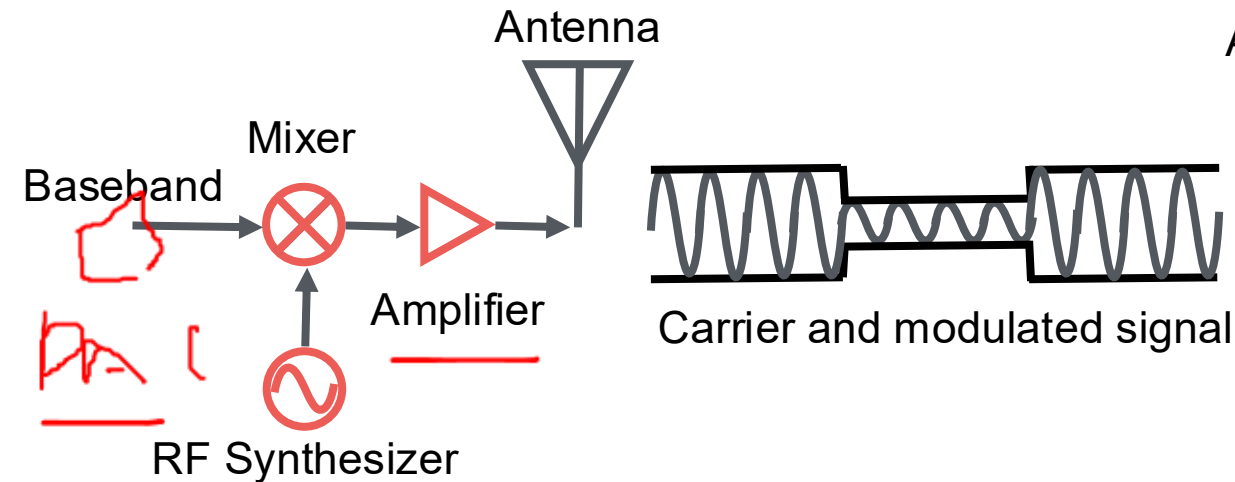
- Wireless protocols require power and computation



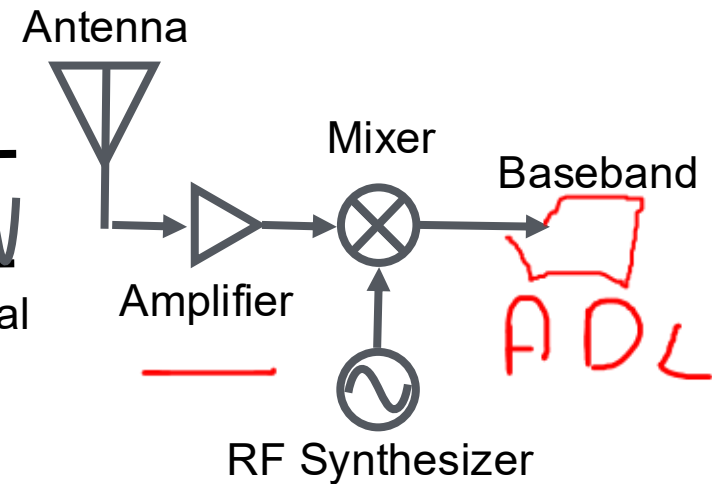
RFIDs can't perform typical wireless functions like carrier sense or rate adaptation

Why are active radios power-hungry?

Active TX



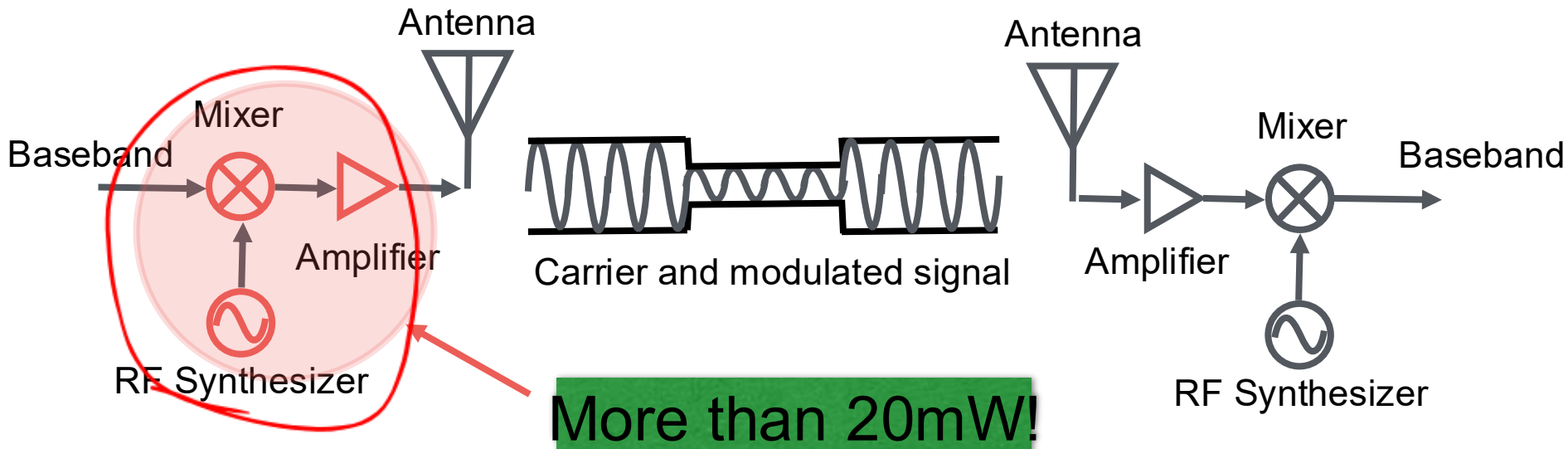
Active RX



Why are active radios power-hungry?

Active TX

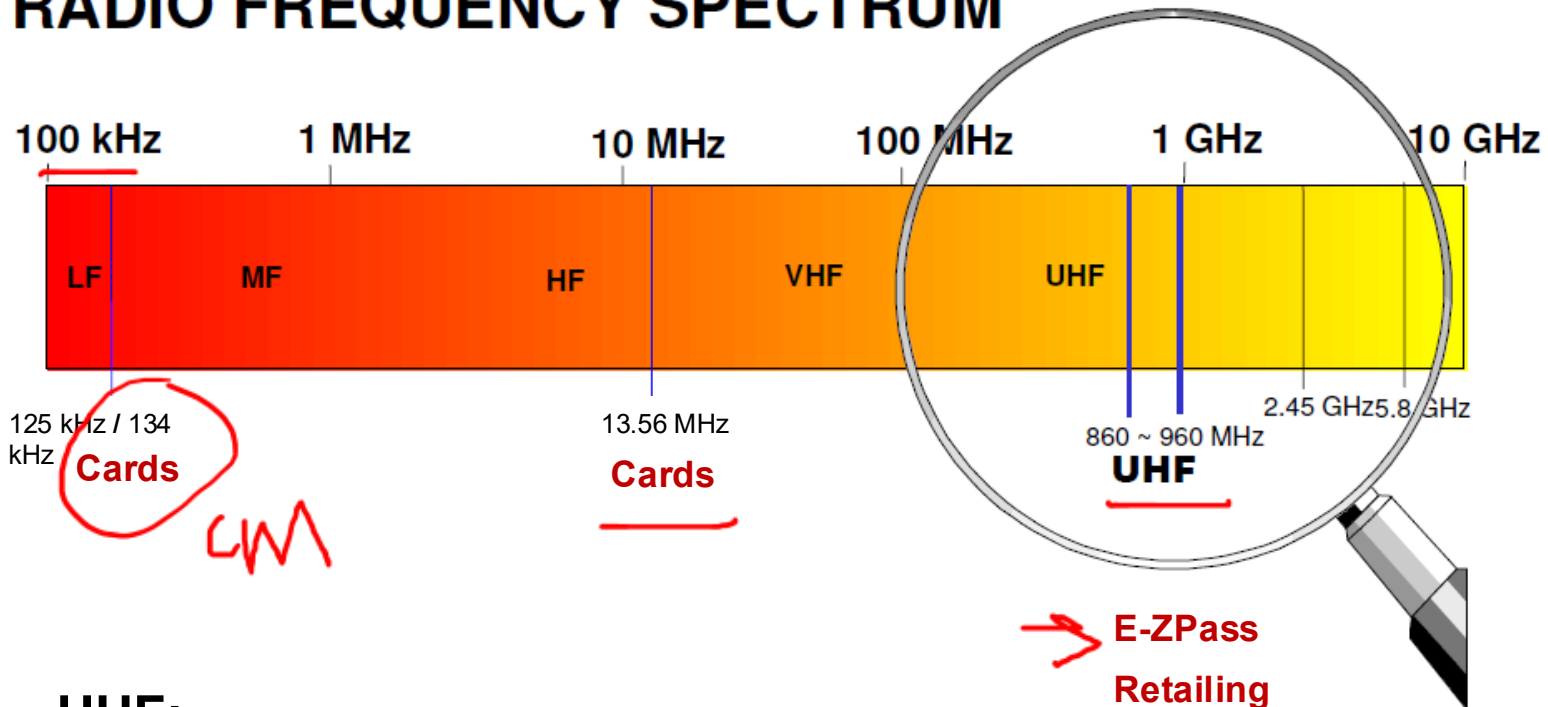
Active RX



RFID Background

ISM band

RADIO FREQUENCY SPECTRUM

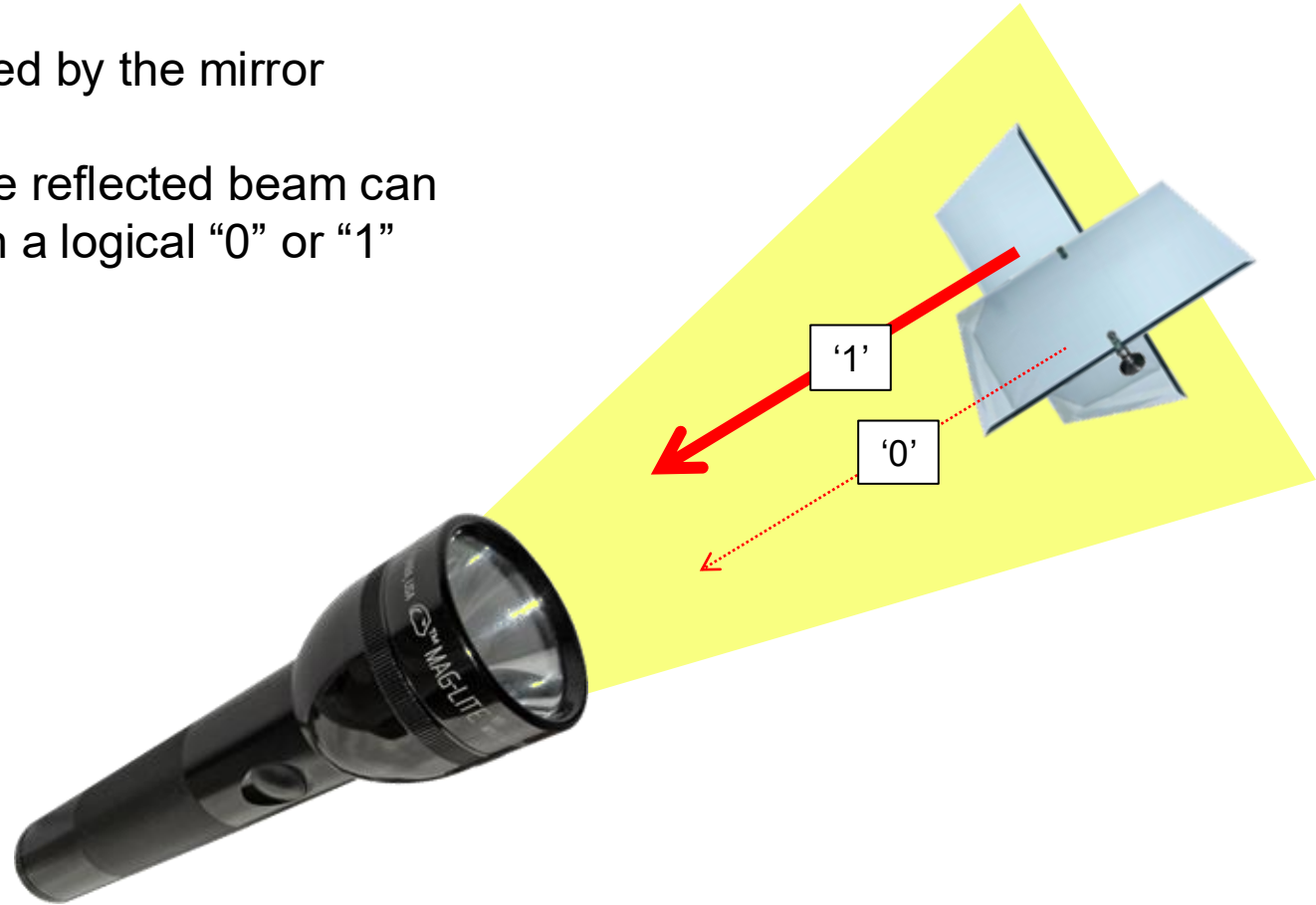


UHF:

- Achieves higher range (few meters v.s. cm)
- Uses backscatter communication instead of inductive coupling

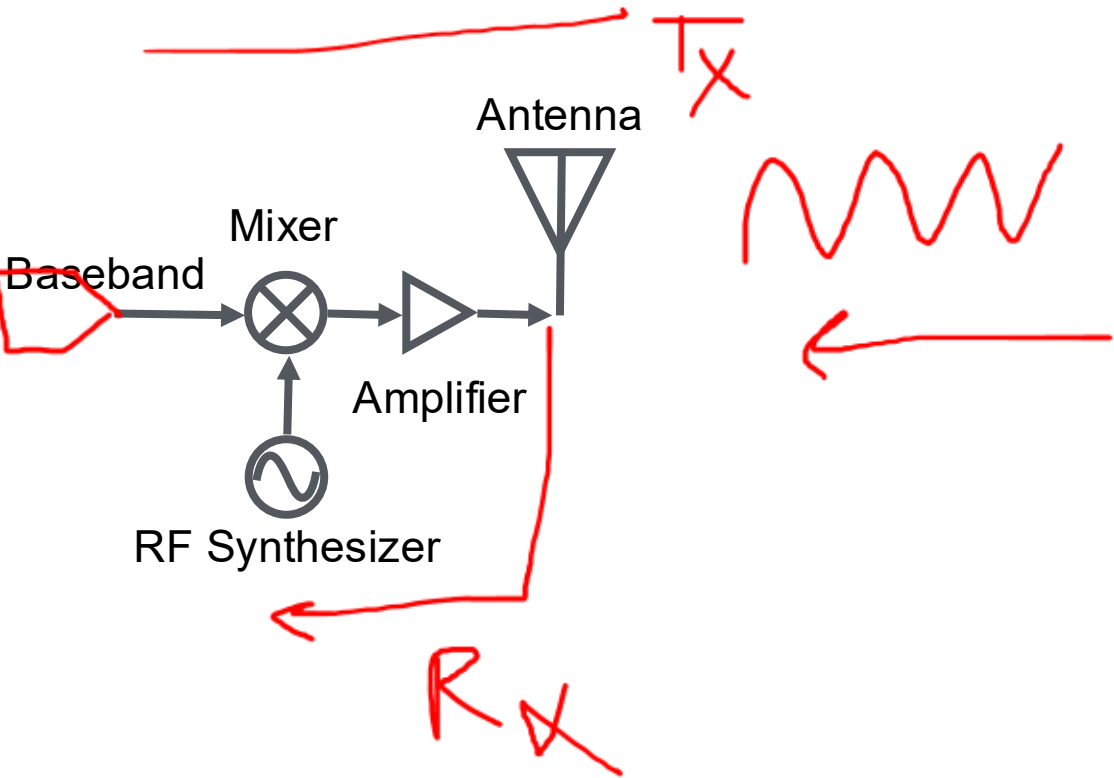
Backscatter Communication

- A flashlight emits a beam of light
- The light is reflected by the mirror
- The intensity of the reflected beam can be associated with a logical “0” or “1”

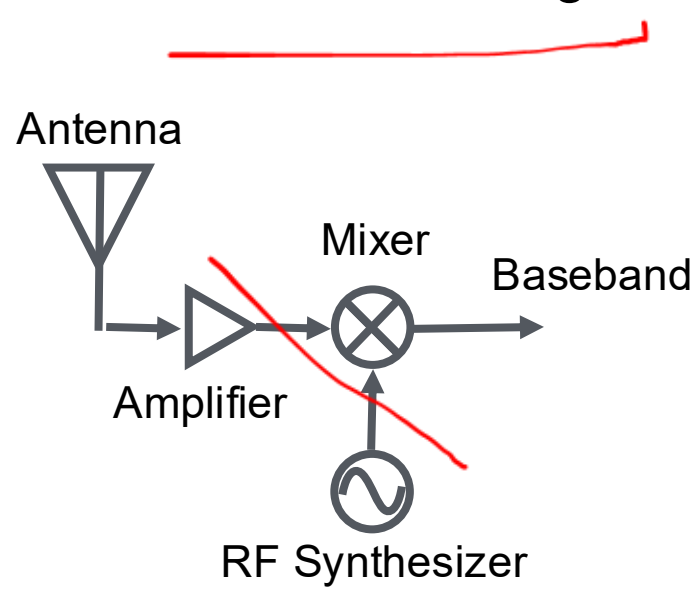


Backscatter enables ultra-low power wireless

Backscatter Reader

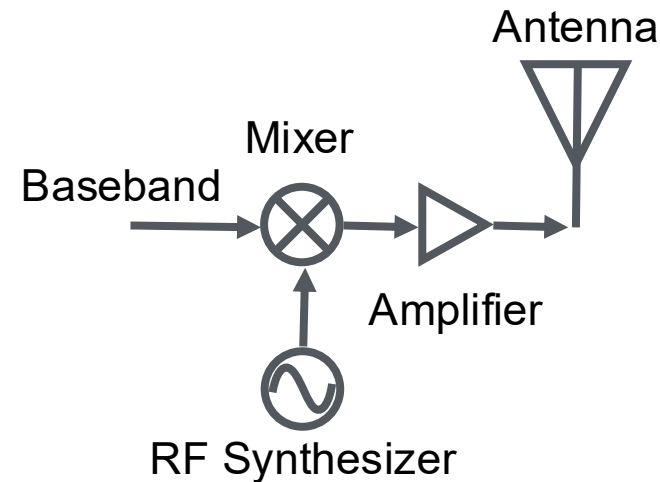


Backscatter Tag

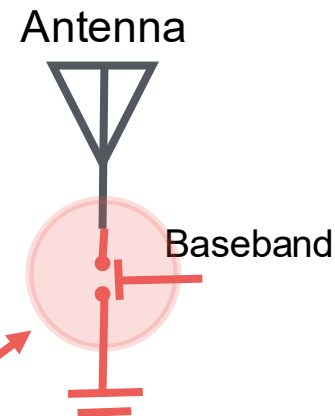


Backscatter enables ultra-low power wireless

Backscatter Reader



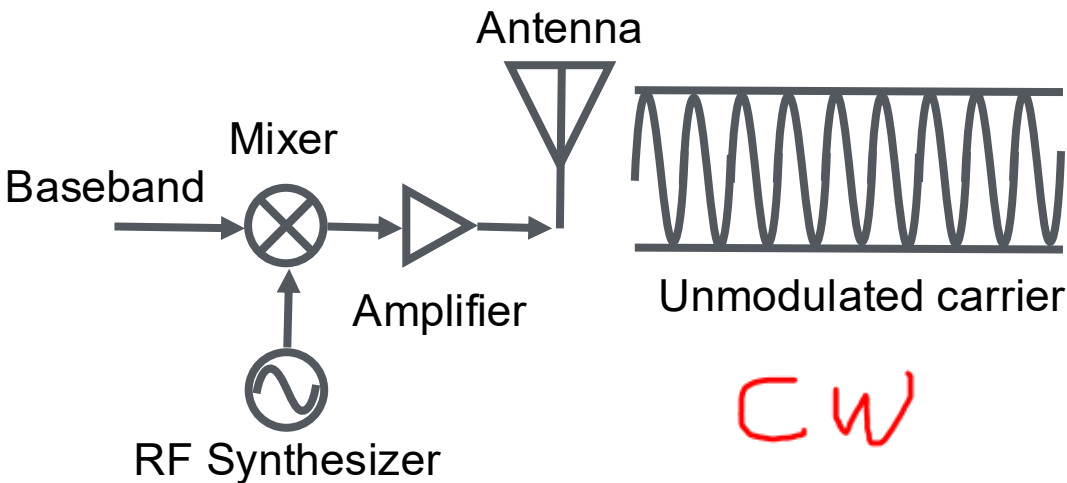
Backscatter Tag



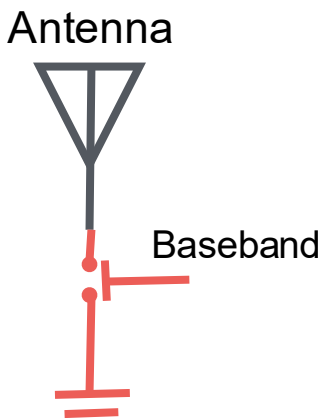
Only one transistor

Backscatter enables ultra-low power wireless

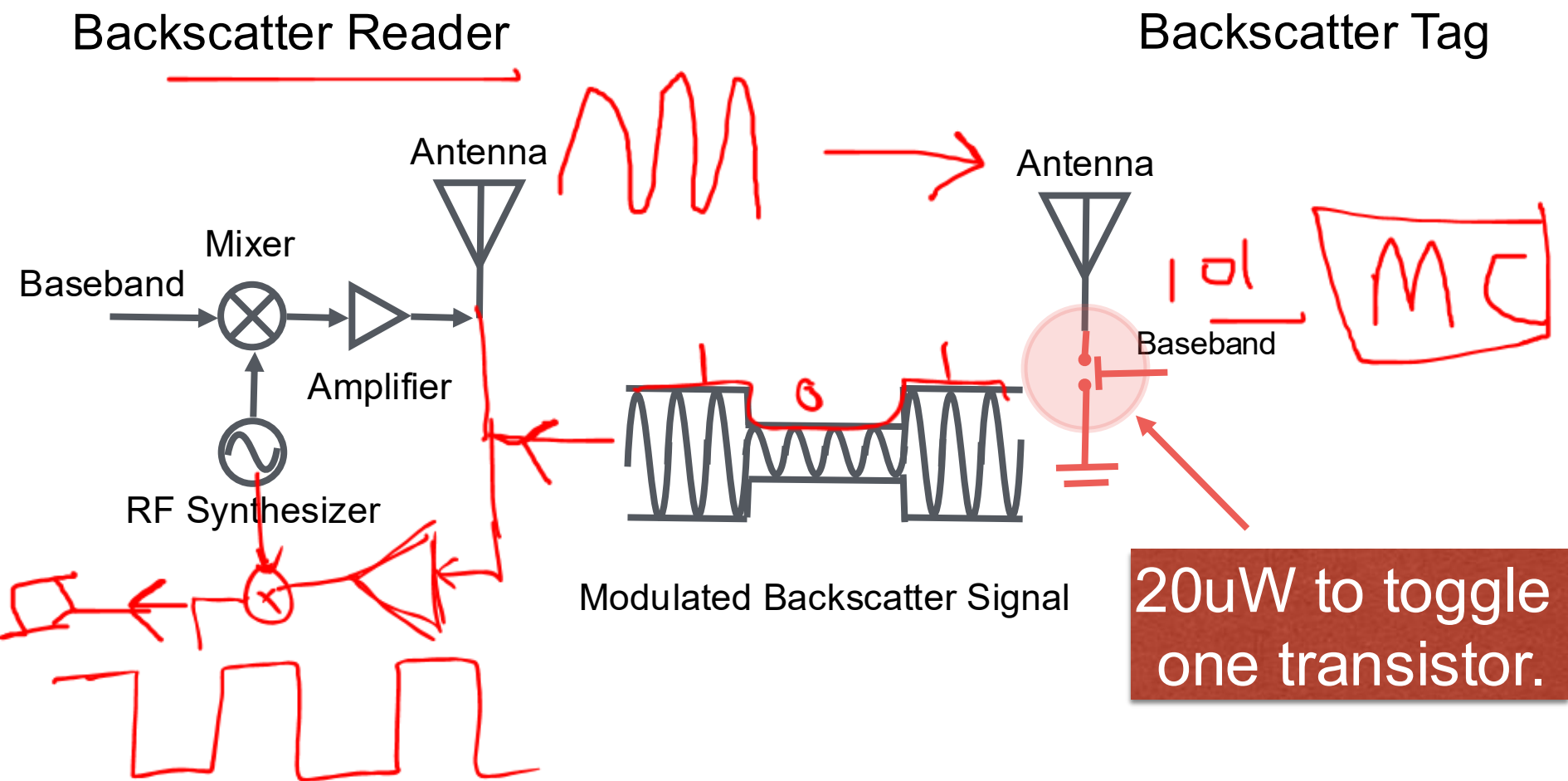
Backscatter Reader



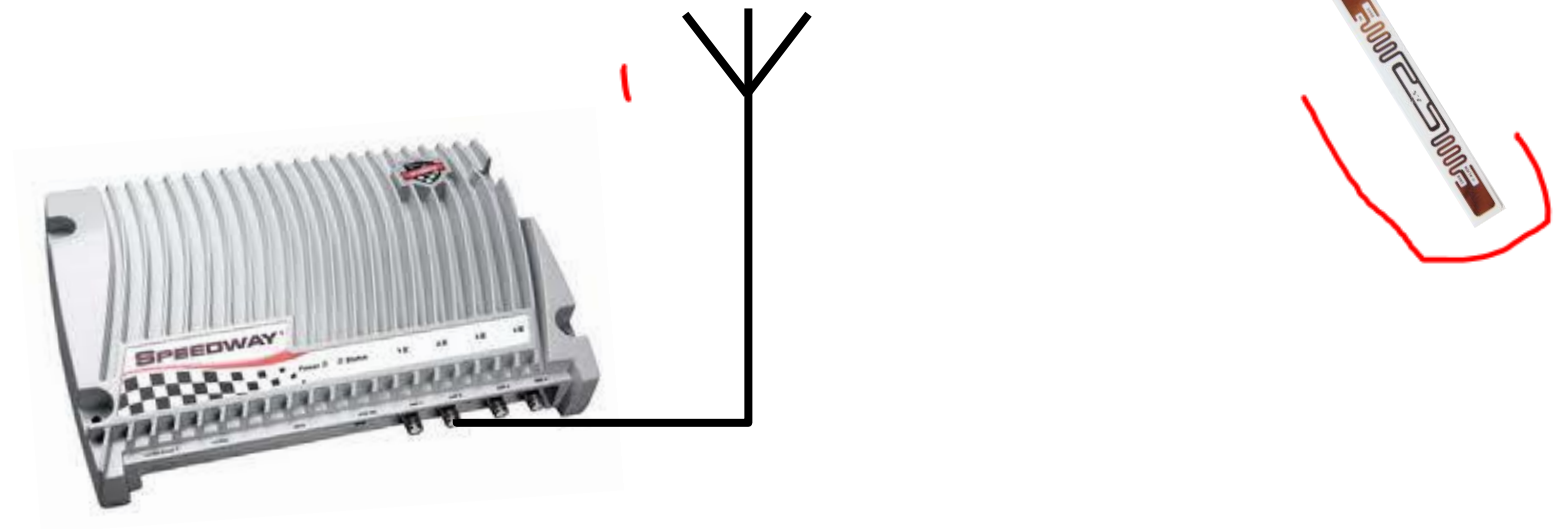
Backscatter Tag



Backscatter enables ultra-low power wireless



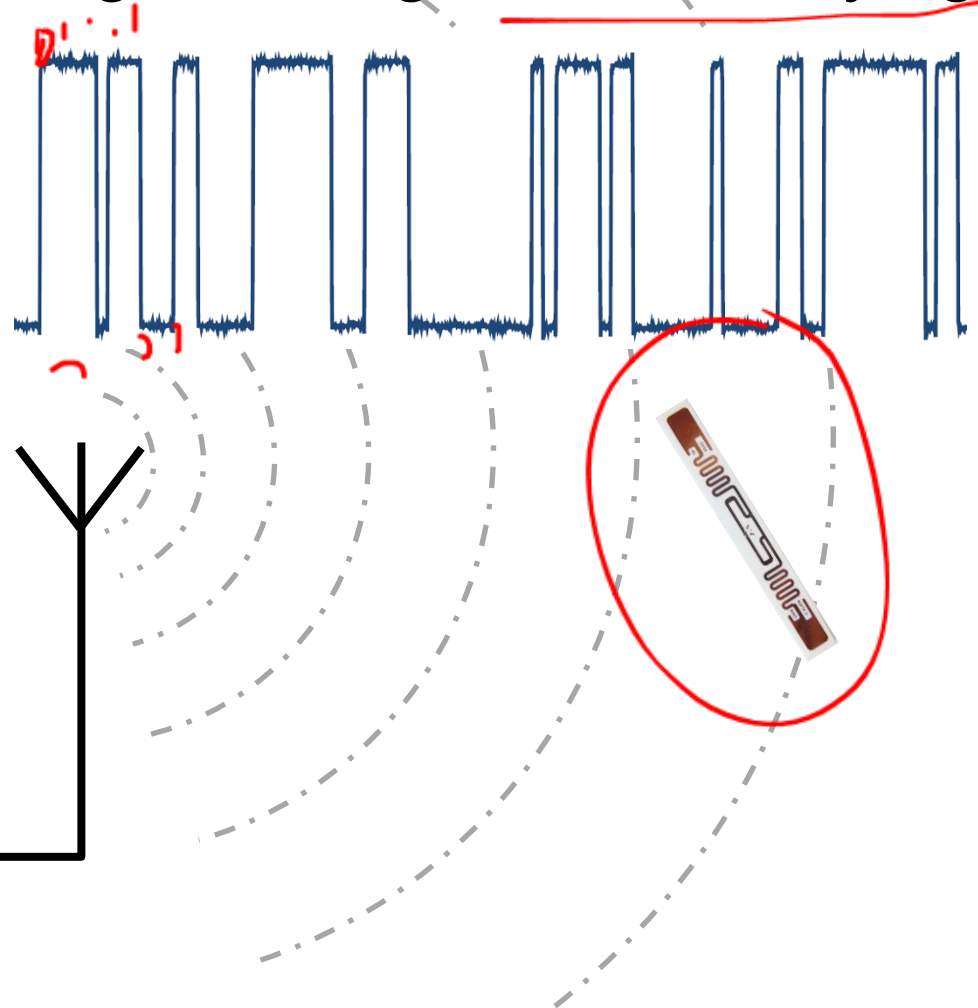
Backscatter Communication



Backscatter Communication

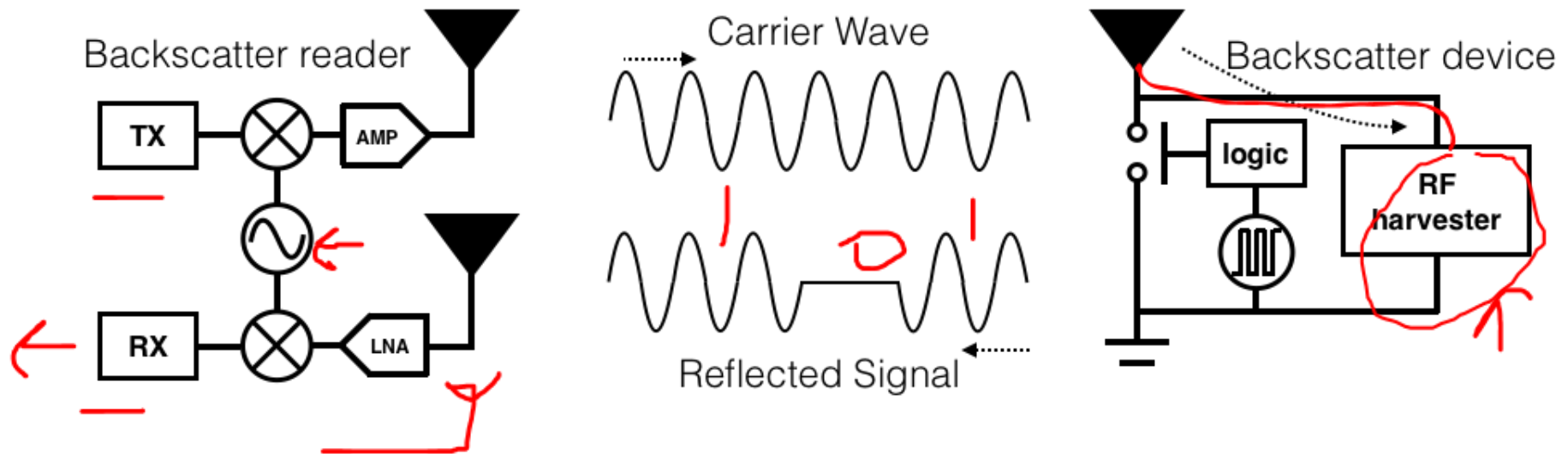
Tag reflects the reader's signal using ON-OFF keying

Reader shines an RF signal on nearby RFIDs



No Carrier Frequency offset

Backscatter — An Ultra Low Power Communication Primitive

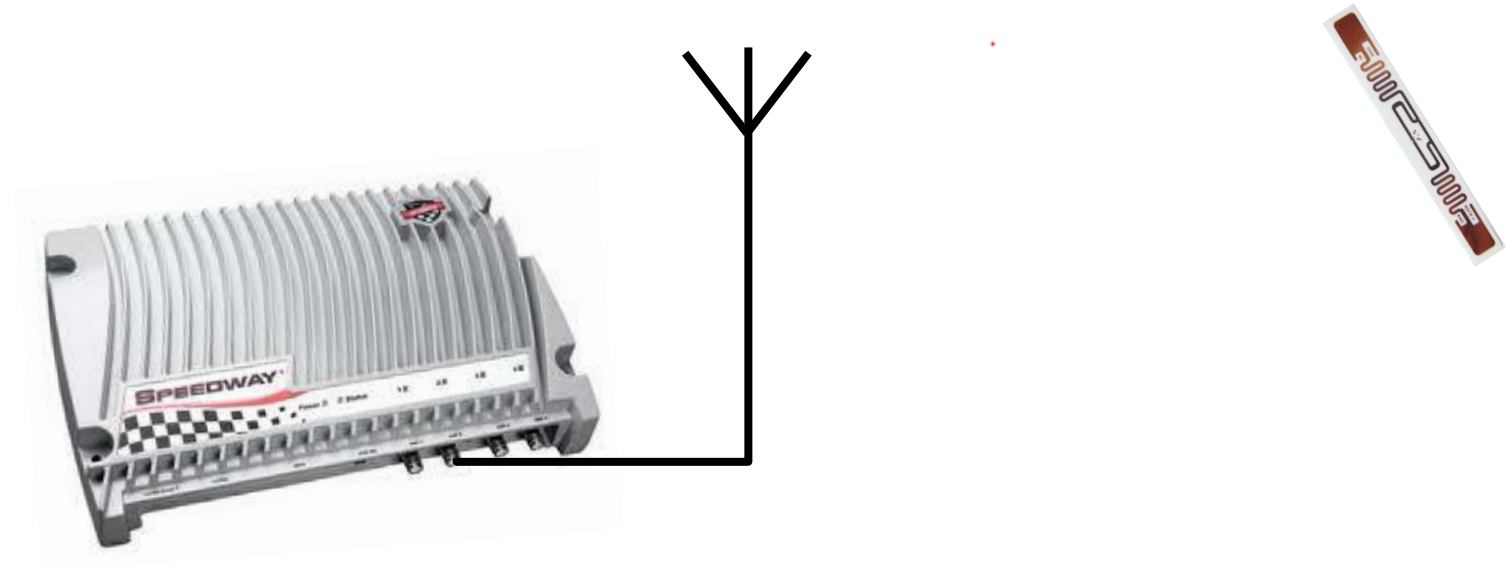


Backscatter enables ultra low-power wireless communication.

Backscatter Communication

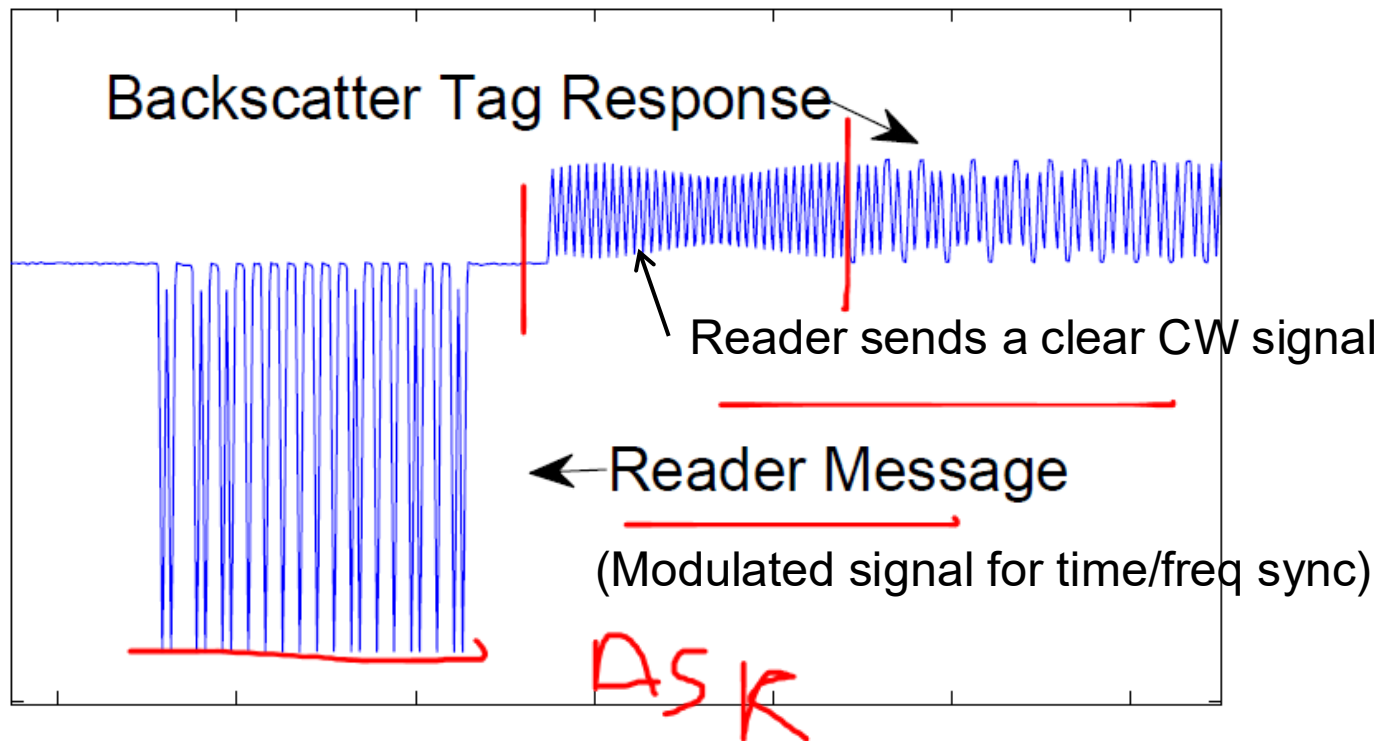
RFIDs are synced by the reader's signal:

- Time synchronization ($\sim .1\mu\text{sec}$)
- Frequency synchronization



Challenges of Backscatter

Time and Frequency synchronization



Challenges of Backscatter

RFIDs cannot hear each other

- Collisions
- Slotted Aloha protocol (next class)

Cannot adapt modulation to channel quality

- Don't exploit a good channel to send more bits per symbol
- Don't react to a bad channel

Research Platforms

- RFID tag: passive stickers

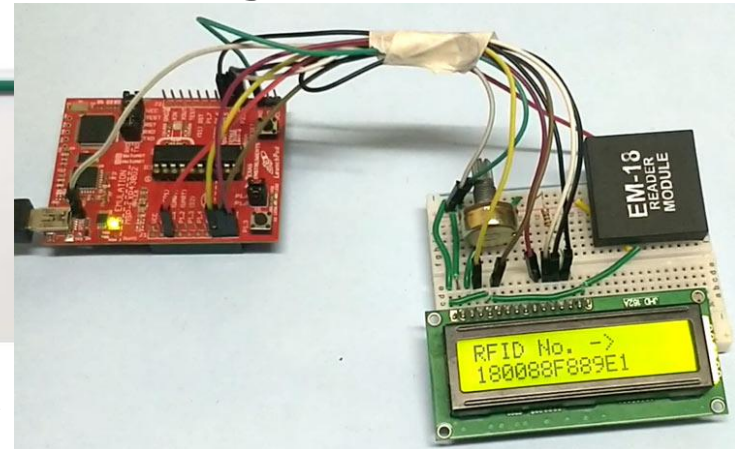


Research Platforms

- RFID tag: computational RFIDs



- MSP430 Microcontroller
 - 8KB RAM + 116KB Flash + 12 bit ADC/DAC
- Sensors:
 - Accelerometer + temperature + voltage + external sensors



RFID Project Ideas: [RFID Interfacing with MSP430 Launchpad](#)

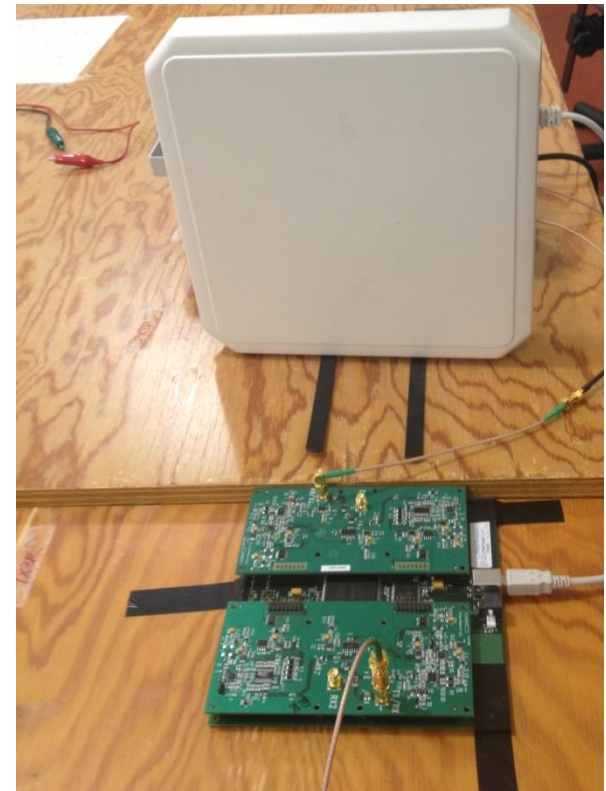
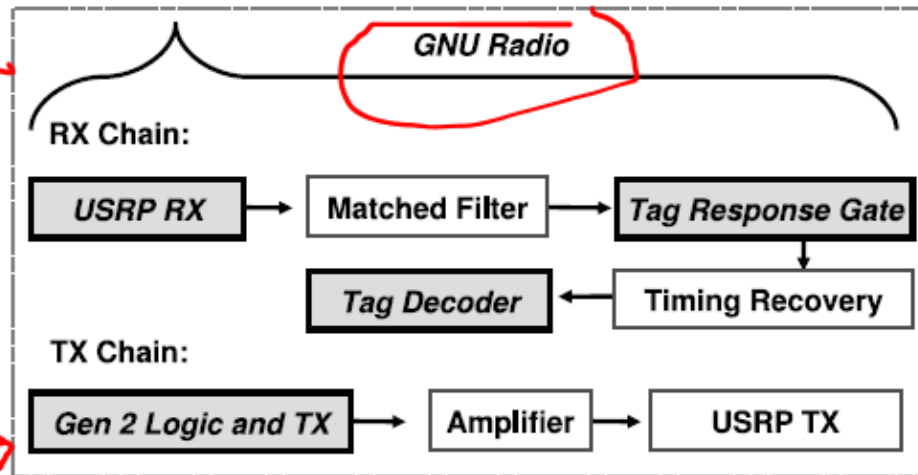
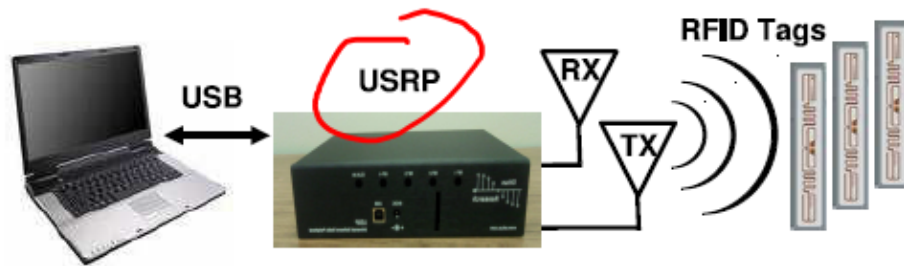
Research Platforms

- Reader: Think magic/Impinj

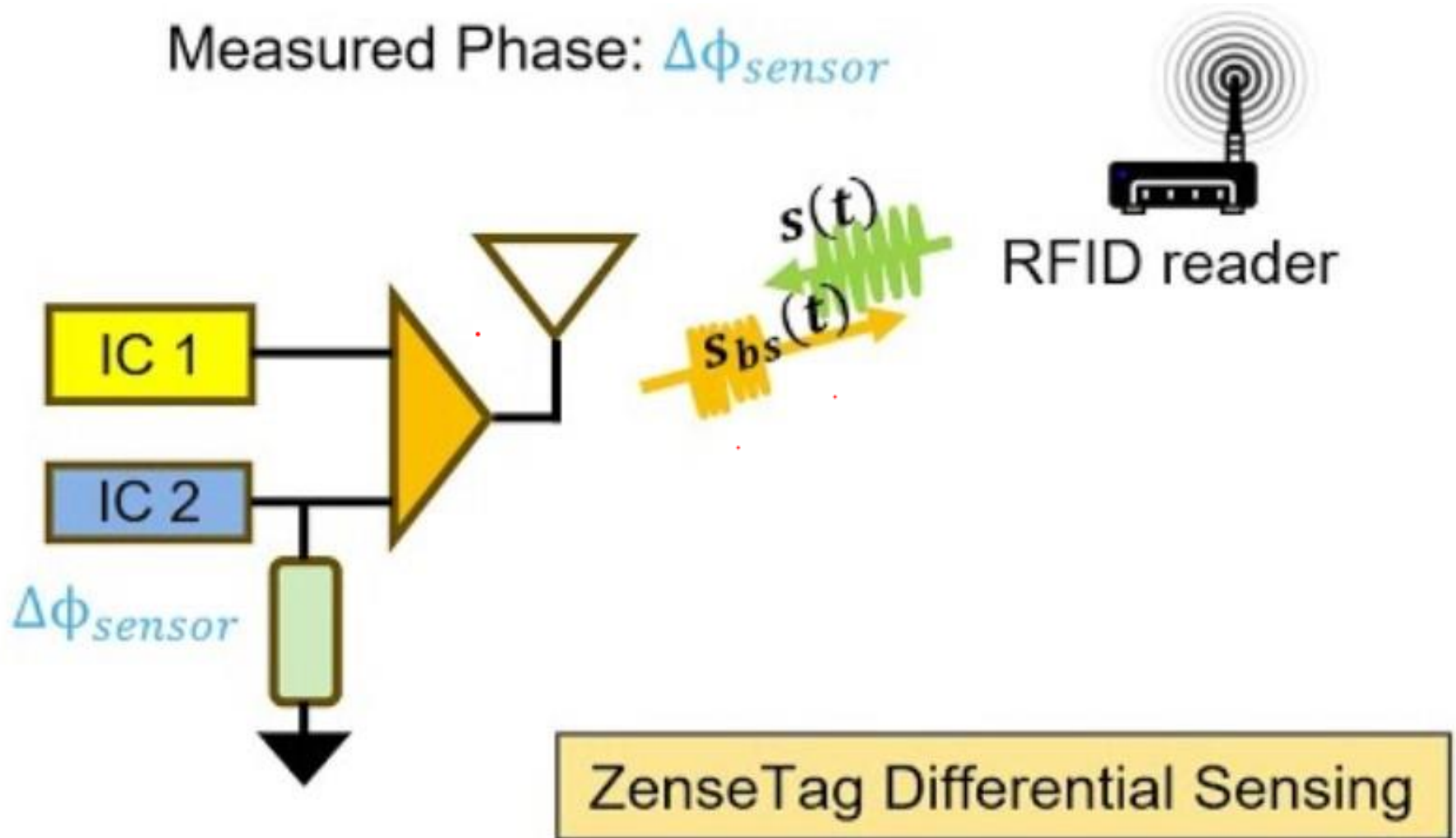


Research Platforms

- Reader: software radio based Gen-2 reader



Research Projects



Conclusion on RFID

- Many applications for low power networks.
- Nodes need to be very simple (low cost, low power) → cannot have advanced functionalities
- Need new research ideas that can enable advanced protocol.
- What would you do if I give you so many RFIDs?

PHY Layer Techniques: Outline

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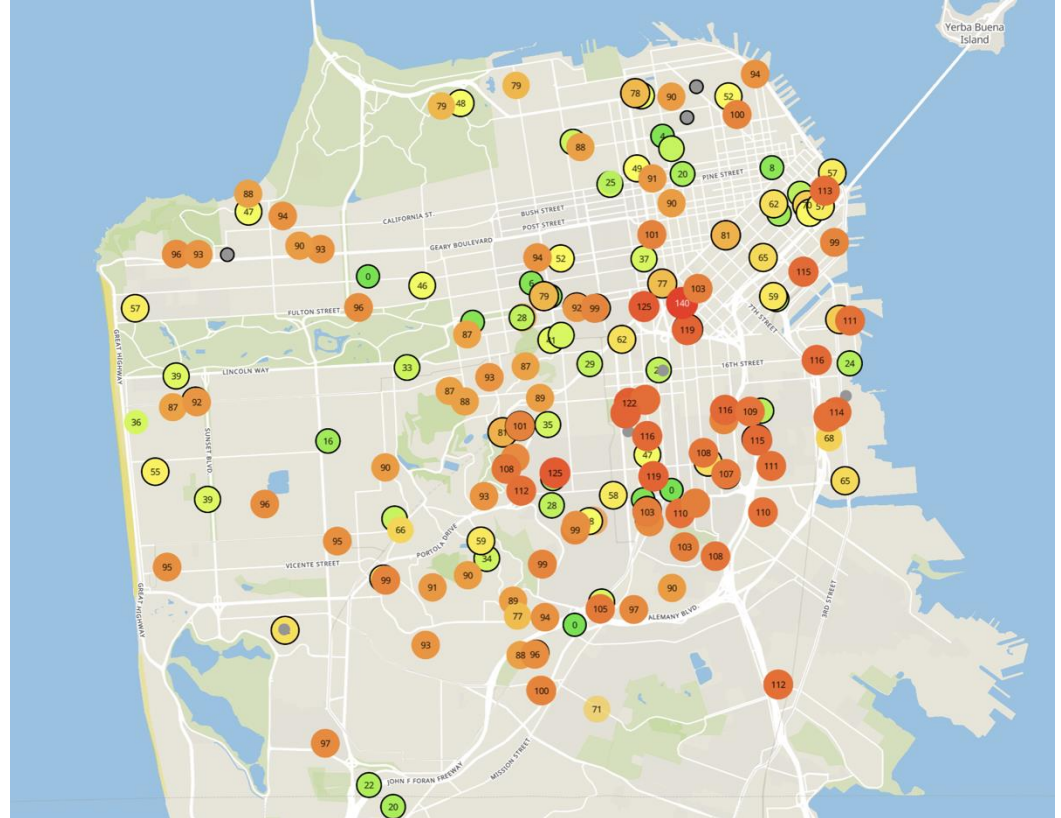
Example application: air quality monitoring



[1]



[2]

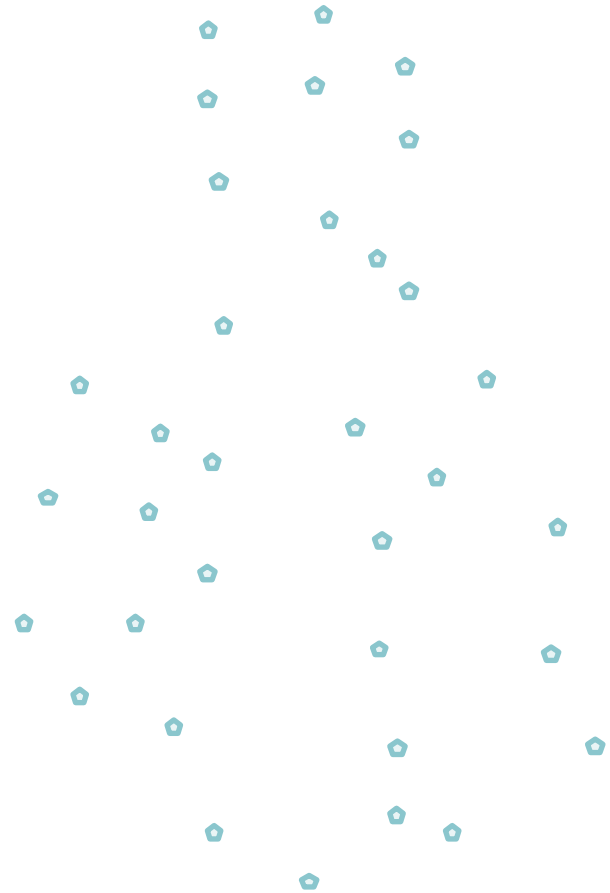


[2]

[1] Cheng et al. AirCloud: a cloud-based air-quality monitoring system for everyone. 2014. [2] Purple Air. 2019.

How do we collect data from MANY sensors?

- Manually collect measurements
- Connect it to WiFi (or Ethernet)
- Pay for cellular access



2.4GHz protocols mostly used for indoor applications

- Range for the wireless protocols we've looked at so far (best case)
 - **BLE:** 100m (~300ft)
 - **802.15.4:** 100m (~300ft)
 - **NFC:** 20cm (~1ft)

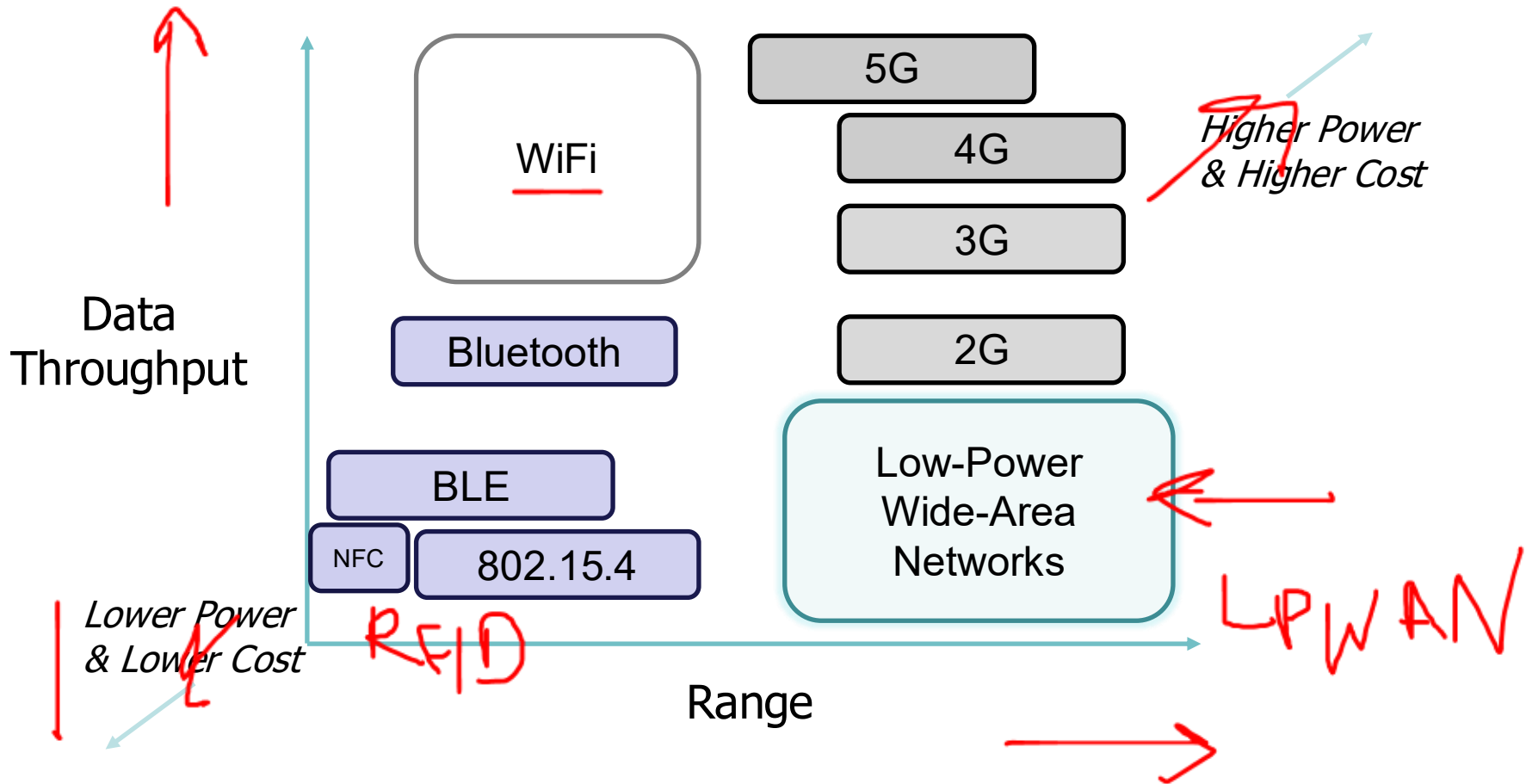
We need another network option

- What requirements do we have?
 - Wide area of coverage
 - Deploy fewer gateways
 - Low power
 - So we can deploy on batteries
- What requirements do we NOT have?
 - Doesn't need high throughput
 - Sensor data is relatively small

Wide area networks

- Communication at the region/city scale rather than the building/residence scale
 - Throughout cities
 - Agricultural deployments
 - Industrial facilities
- City-scale sensing is one very popular domain
 - What might we want to sense throughout a city?

Long-range, low-data needs haven't historically been met



LPWANs overview (common qualities)

- Unlicensed 915 MHz band (902-928 MHz)
- Higher power transmissions: ~20 dBm (100 mW)
- Low data rate: 100 kbps or less
- Range on the order of multiple kilometers
- Collisions? Simple ALOHA access control

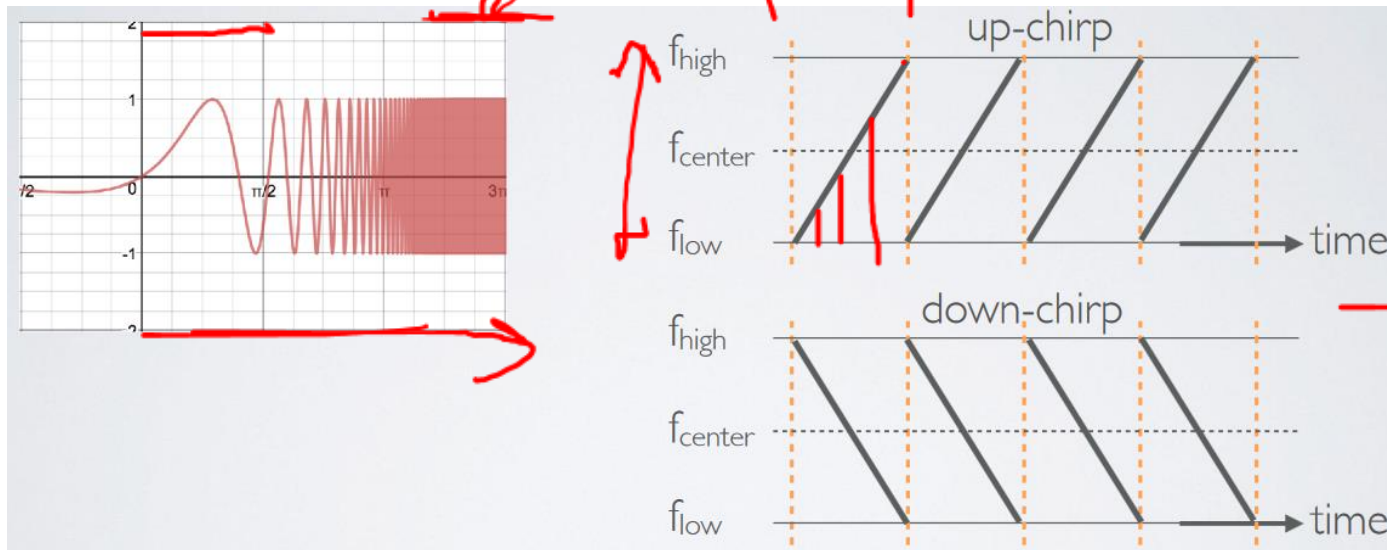
LoRa (Long Range)

- Proprietary LoRa physical layer
 - Owned by Semtech
- Low data rate (1-20 kbps) and long range (~5 km)
- Most popular LPWAN protocol
 - Target of academic research
 - Industry involvement in hardware and deployments

LoRa PHY uses a spread spectrum modulation technique

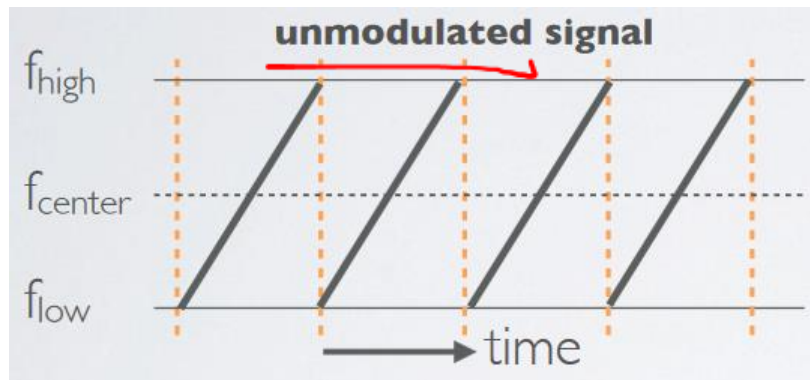
■ Chirp Spread Spectrum (CSS)

- Modulation technique where frequency is varied linearly from (to) lowest to (from) highest within a channel



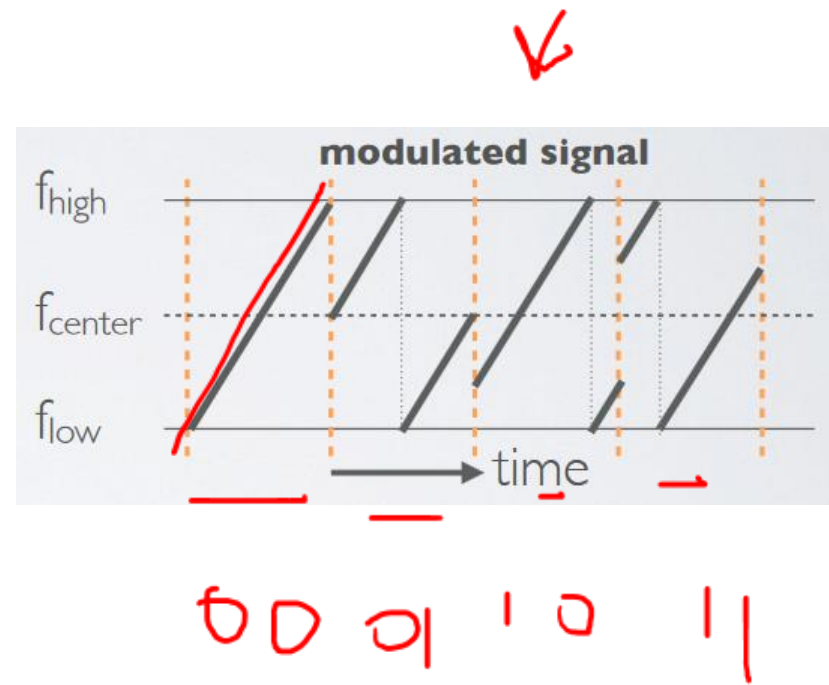
Data Modulation in Chirp Spread Spectrum

- Data is modulated in the starting and ending points of chirp
 - Frequency increases linearly, modulo bounds of the channel



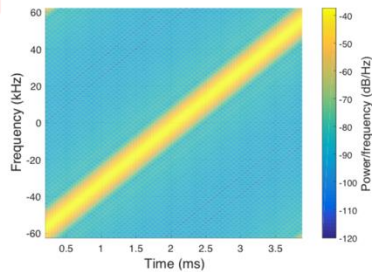
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Input Binary Data

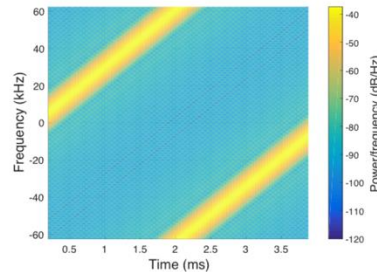


Data Modulation in Chirp Spread Spectrum

 LoRaWAN™ : 1-bit encoding



'0'

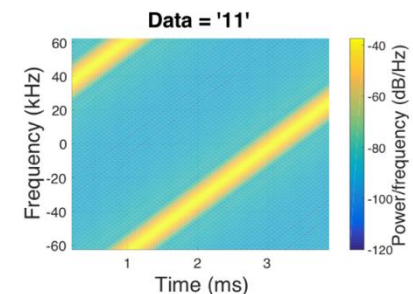
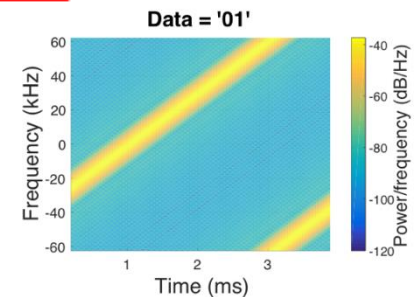
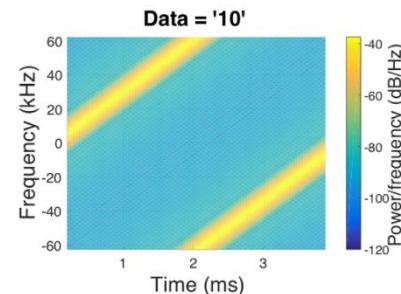
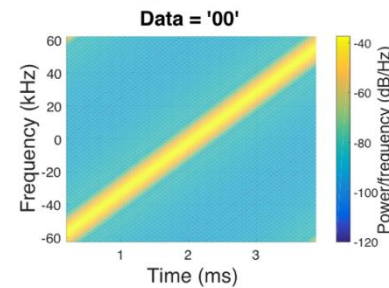


'1'

In general,
 n bits $\Rightarrow$ divide BW to 2^n initial frequencies

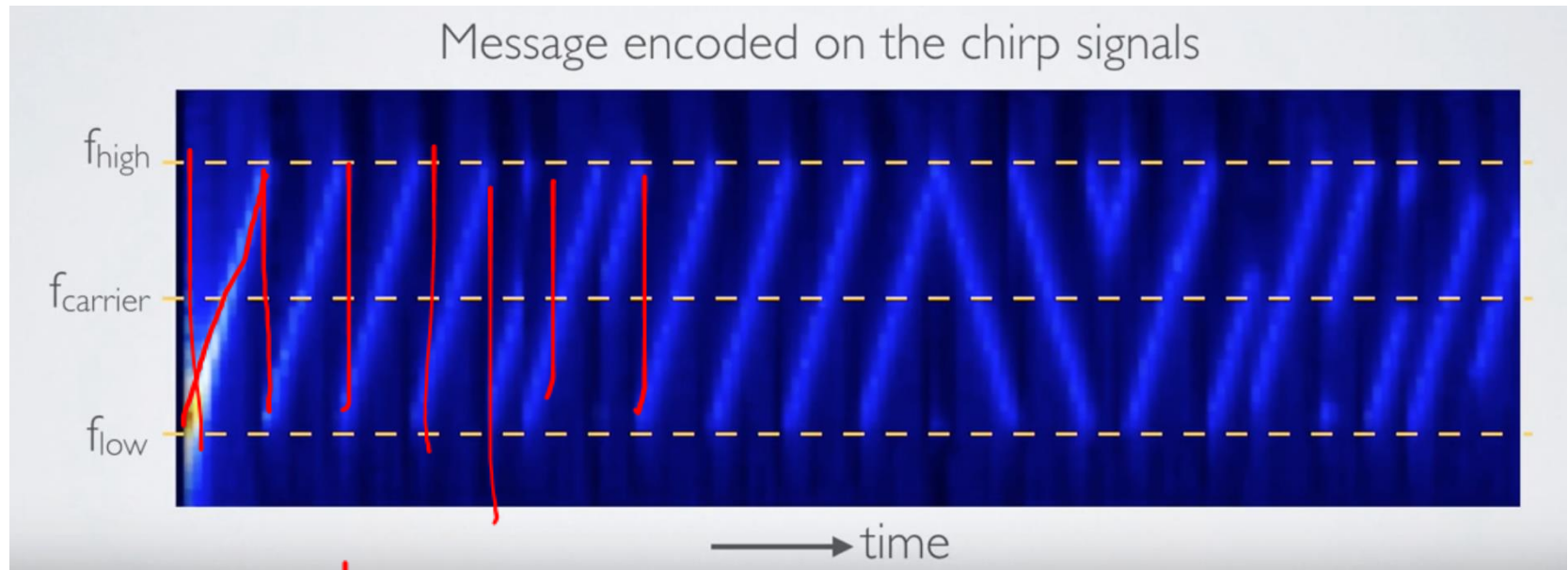
Remember, No. of symbols = 2^n

 LoRaWAN™ : 2-bit encoding



Source: <https://swarunkumar.com/slides/choir-sigcomm2017.pdf>

LoRa Modulated Signal

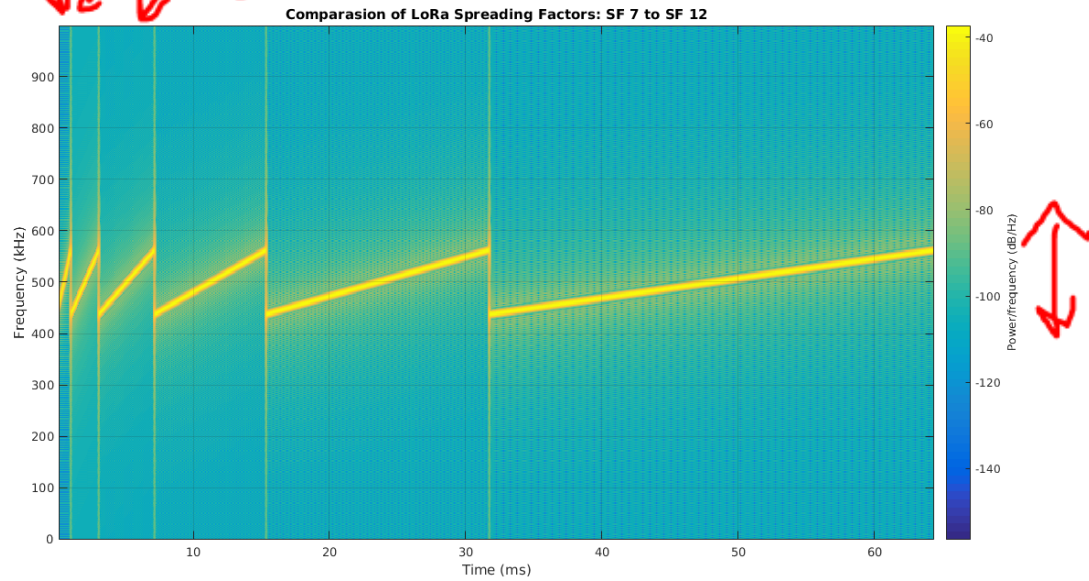


Source: <https://youtu.be/lg0eZWZFKiE>

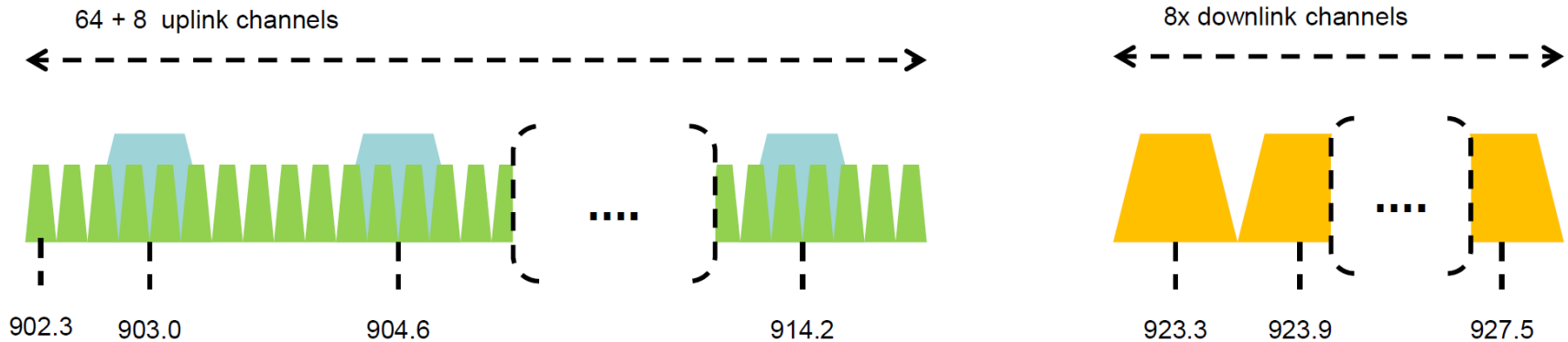
CSS has a Spreading Factor which determines bit rate

- Spreading Factor is essentially the rate-of-change of frequency (sweep rate)
 - Slope of the line
 - Lower values of spreading factor (steeper slope) are faster data rate
- Important: different spreading factors are (mostly) orthogonal!
 - Two can overlap in time, space, and channel without a collision

LoRa has 6 spreading factors:
SF7 to SF12



LoRa channels (in the US)



- Sixty-four, 125 kHz uplink channels
 - Frequency Hopping over the 64 uplink channels
 - Plus eight, 500 kHz overlapping uplink channels (not used in practice)
- Eight, 500 kHz downlink channels

LoRa gateways

- No synchronization with end devices
- Instead listen to entire bandwidth simultaneously
 - Only 12 MHz total
 - Recognize preambles and allocate hardware to decode packet
 - Cheap gateways: 8 decoders
 - Good gateways: 64 decoders

LoRa data rates

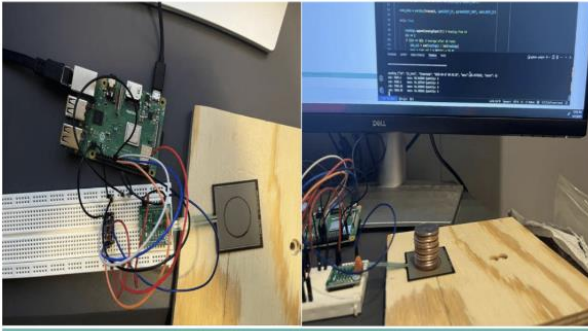
- Data rate options depend on channel in use
 - Unbalanced uplink and downlink
- 64-channel uplink
 - 1-5 kbps data rate
- Allowable rates based on US dwell time restriction (400 ms)
 - Different in different regions

Data Rate Index	Spreading Factor	Bit Rate
<i>125 kHz Uplink Rates</i>		
0	SF 10, 125 kHz	980 bps
1	SF 9, 125 kHz	1760 bps
2	SF 8, 125 kHz	3125 bps
3	SF 7, 125 kHz	5470 bps
<i>500 kHz Uplink Rates</i>		
4	SF 8, 500 kHz	12500 bps
<i>500 kHz Downlink Rates</i>		
8	SF 12, 500 kHz	980 bps
9	SF 11, 500 kHz	1760 bps
10	SF 10, 500 kHz	3900 bps
11	SF 9, 500 kHz	7000 bps
12	SF 8, 500 kHz	12500 bps
13	SF 7, 500 kHz	21900 bps

LoRa link budget

- Typical TX power 20 dBm
 - Up to 30 dBm for 64-channel hopping
 - Up to 26 dBm for 8-channel hopping
- Receive sensitivity -119 dBm
 - **Compare to -100 dBm for 802.15.4 and -95 dBm for BLE**
- Resulting range is about a kilometer in urban environments

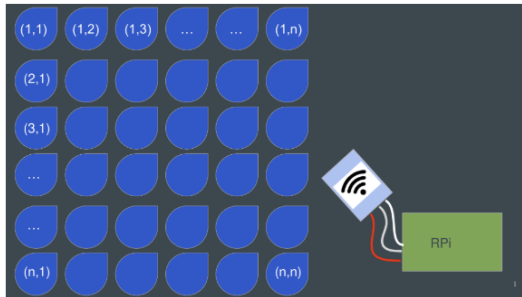
IoT projects on RFID and LoRa



Warehouse Inventory Tracker

Josh Christensen and James Laun

Spring 2025



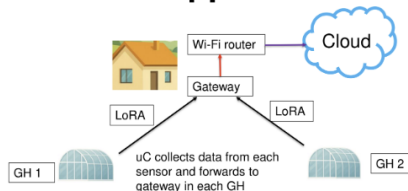
RF Grid Movement Tracker

Gabriela Crother-Collado and Joseph Marino

Spring 2025



Our Approach



Greenhouse Monitoring with LoRa

Raj Bakhunchhe and Matthew Fiorenza

Spring 2025

